Worksheet # 1 Part A - Multiple Choice
Calculus Physics 2

ame: Key Date: _____

Questions 1 though 10

1) What quantities are required to completely describe the Physical quantity that is a scalar?

(A) Magnitude (with unit) only
B) Direction only
C) Both magnitude and direction

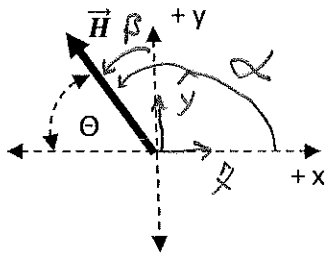
2) The projection of a vector along one of the principle axes is

A) the unit vector.
B) the magnitude of the vector.
C) the direction of the vector.
(D) the scalar component of the vector along that axis.
E) None of the above

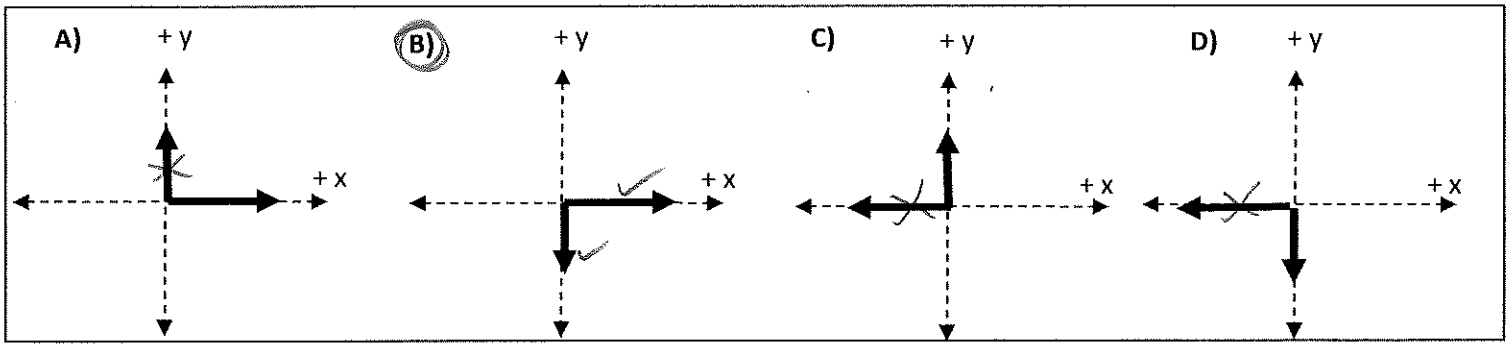
3) Vector $\vec{\mathbf{H}}$ is shown in the graph, what are the angles α and β for this vector if Θ = 50°?

$\alpha = 180° - \Theta = 130°$

$\beta = 90° - \Theta = 40°$



4) Which graph below best represents the vector $\vec{\mathbf{C}} = (8\ N)\hat{\mathbf{x}} + (-4\ N)\hat{\mathbf{y}}$?



5) When the Dot product of two vectors (non-zero vectors) is zero, what is the angle between the vectors?

(A) 90°
B) 0°
C) 180°
D) 45°
E) Both 0° and 90°

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos(\phi)$ ← angle between vectors

$\cos(90°) = 0$

6) Sphere 1 has 3 times the charge as Sphere 2, how does the magnitude of the force that q_1 exerts on q_2 compared to the force that q_2 exerts on q_1?

A) half as much.
B) four times as much.
C) twice as much.
(D) the same.



$|\vec{F}_{1on2}| = |\vec{F}_{2on1}|$

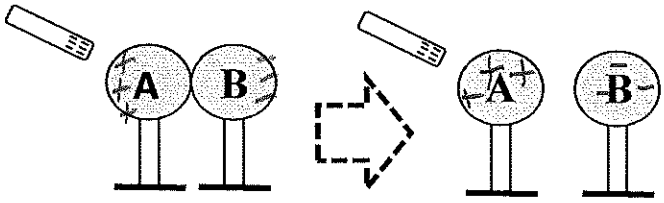
7) What physical process enables a balloon that is charged to stick to a wall that is not charged?

A) Magnetic induction
B) Electric potential
(C) Electrostatic polarization
D) Gauss' law
E) Maxwell's law



8) A negatively charged rod is brought near but does not touch two neutral conducting sphere labeled A and B that are in contact. The rod is kept in place, but the spheres are separated. What if any is the net charge on the spheres when they are separated?

(A) Sphere A is positive & Sphere B is negative.
B) Sphere A is negative & Sphere B is positive.
C) Sphere A has no net charge & Sphere B is negative.
D) Sphere A is positive & Sphere B has no net charge.
E) Sphere A and Sphere B have no net charge.

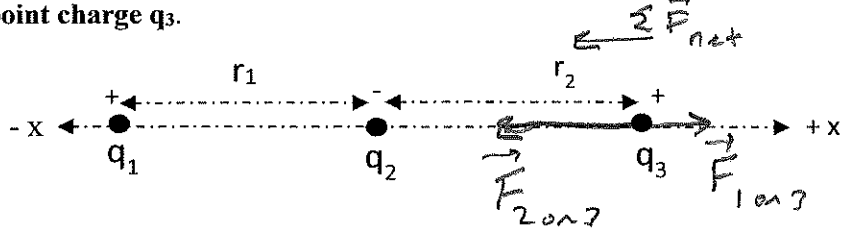


9) In Electrostatics, if two objects repel each other must both objects have a net charge?

A) No, there can also be ground induced charge.
B) No, there can also be polarization induced charge.
C) No, there can also be a force due to the electric field.
(D) Yes
E) Yes, but only if they are different types of charge.

10) Three point charges are placed on the *x*-axis with the same magnitude of net charge, but different types shown below. The separation between q_1 and q_2 is equal to the separation between q_2 and q_3. Determine the direction of the net electrostatic force exerted on **point charge q_3**.

q_1 is positive
q_2 is negative
q_3 is positive
$r_1 = r_2$



A) The net electrostatic force is in the positive x direction.
(B) The net electrostatic force is in the negative x direction.
C) The net electrostatic force is zero on q_3.
D) Can not be determined with information given.

$|\vec{F}_{2on3}| > |\vec{F}_{1on3}|$

$r = r_2$ $\quad r = r_1 + r_2$